

Number of antinodes	2	3	4	5
Frequency of oscillator	101 Hz	149 Hz	201 Hz	251 Hz

The data in the table above were recorded during an experiment in which a standing wave pattern was produced on a stretched string of length 0.5 m. One end of the string was attached to a fixed wall and the other was connected to an oscillator. The speed of the waves on the string is most nearly

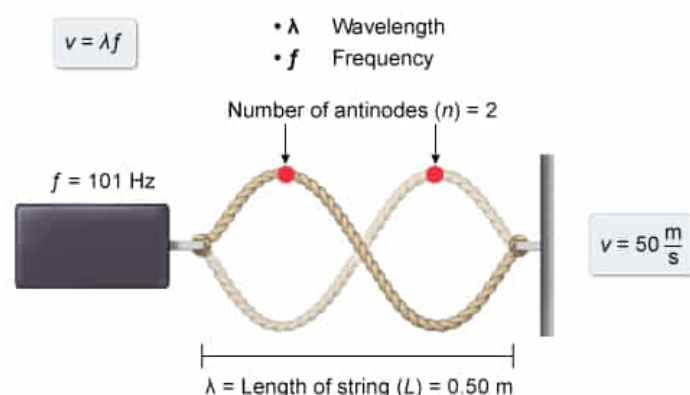
- ✓ ☒ A. 50 m/s [48%]
☐ B. 100 m/s [32%]
☐ C. 150 m/s [11%]
☐ D. 200 m/s [7%]

Correct

48% answered correctly

Explanation:

Wave speed (v) for standing wave configurations



When a wave is restricted to fixed region by a boundary (eg, a string with one end fixed to a wall and the other connected to an oscillator), a wave can form that appears to stand in place and not travel horizontally. These **standing waves** are created at a certain set of frequencies called **harmonics**, where incident and reflected waves perfectly overlap each other.

The **wave speed** v is a property of the medium and determined by the product of the **wavelength** (λ) and **frequency** (f):

$$v = \lambda f$$

The λ of a standing wave is determined by the length of the string L and the number n of **nodes or antinodes**. At the fundamental frequency, there is one antinode present ($n = 1$) that oscillates between nodes located at each boundary. Therefore, L is half the wavelength, or $\lambda = 2L$. Larger harmonic frequencies correspond to a greater number of antinodes and shorter wavelengths, as given by the formula:

$$\lambda_n = \frac{2L}{n}$$

In the question, the oscillator frequency associated with $n = 2$ is $f = 101$ Hz. Because $\lambda = L = 0.5$ m when $n = 2$,

$$v = 101 \text{ Hz} \cdot 0.5 \text{ m} \approx 50 \text{ m/s}$$

The data for **higher values of n** give approximately the same result:

$$v = 149 \text{ Hz} \cdot 0.33 \text{ m} \approx 50 \text{ m/s}$$

$$v = 201 \text{ Hz} \cdot 0.25 \text{ m} \approx 50 \text{ m/s}$$

$$v = 251 \text{ Hz} \cdot 0.20 \text{ m} \approx 50 \text{ m/s}$$

Therefore, the speed of the waves is most nearly 50 m/s.

(Choices B and C) These answers correspond to a miscalculation of wavelength, which is defined as the distance from crest to crest or, equivalently, trough to trough.

(Choice D) This answer results from using the formula $v = \frac{f}{\lambda}$, which does not have the correct dimensions for speed.

Educational objective:

The speed of a wave is given by the product of its wavelength and frequency. The wavelength of a standing wave on a string can be determined from the length of the string and the number of antinodes.

Time spent:0 sec

Subject:

Physics

QID:404089

System:

Light and Sound